Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Vision-Language-Action Based Autonomous Task Intelligence for Contact-Rich Robotic Manipulation			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Advanced Robotics		AI	
	Medium Category	Small Category	Medium Category	Small Category
	Intelligent Robotics	Contact-Aware Robotic Intelligence	Multimodal AI	Physical AI/ Vision-Language-Action Learning
Concept of Proposed Technology	The proposed technology focuses on the development of Physical AI based on a Vision–Language–Action (VLA) learning framework, aiming to enable intelligent decision-making in contact-rich environments. While recent Vision–Language–Action models have shown promising results in non-contact and visually driven tasks, their application to contact-rich environments remains limited, as the use of explicit contact and interaction data has not been widely explored. In many real-world robotic applications, task execution is strongly influenced by physical interaction with objects and the surrounding environment. Variations in contact conditions, object tolerance,			

(max 10 pages)

	and environmental uncertainty make it difficult for conventional robotic systems, largely based on predefined motions or rigid control rules, to operate robustly across different scenarios. The proposed VLA-based Physical AI framework integrates three complementary elements. Visual information supports the perception of objects and spatial context relevant to physical interaction. Language is used as a high-level representation of task context and intent, enabling abstraction beyond raw sensor data. Actions are generated as adaptive physical behaviors informed by both perceived context and interaction feedback arising from contact, rather than predefined motion sequences. By unifying perception, contextual understanding, and action, the proposed technology aims to move beyond simple action generation toward context-driven decision-making. This allows AI systems to adapt their behavior when contact, uncertainty, or interaction dynamics play a critical role. The approach is designed to remain independent of specific robot configurations, tools, or industrial processes, enabling broad applicability across diverse robotic platforms.
Rationale for Joint R&D Project	Despite significant advances in robotics and AI, automation in contact-rich environments remains limited. Many industrial robotic systems still require extensive manual configuration or rely on narrowly defined task conditions, resulting in reduced flexibility when physical interaction changes. Joint R&D with Korean SMEs provides an opportunity to bridge advanced AI research with real-world industrial applications. Korean SMEs possess strong expertise in robotic systems, manufacturing processes, and application-specific knowledge, while the proposed technology contributes an AI-centric framework capable of adapting to varying interaction conditions. Through collaboration, the proposed Physical AI technology can be explored, validated, and refined across diverse application scenarios, accelerating its readiness for practical deployment and commercialization while maintaining a flexible, general-purpose AI foundation.
Key Words	1. Vision-Language-Action Learning 2. Multimodal AI 3. Physical AI 4. Robotic Intelligence 5. Contact-Rich Environment

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (4)	**End of the Project**	Level (7)
R&D Objectives	• To develop a Vision–Language–Action based AI framework suitable for contact-rich environments • To enable AI systems to interpret task context and interaction conditions for adaptive decision-making • To demonstrate the general applicability of the technology across different robotic platforms and industrial scenarios			
R&D Target Specifications	• Robust task execution under varying contact and interaction conditions • Reduced dependence on predefined motion rules or rigid task configurations • Applicability across multiple robotic embodiments without requiring system-specific redesign			
R&D Plan	• Conceptual design of a Vision–Language–Action based learning framework for contact-aware decision-making • Exploration of multimodal learning strategies that integrate perception, task context, and adaptive action • Collaborative validation of the technology in representative robotic application scenarios • Refinement of the framework to support broader adoption and future commercialization			

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Development of the core Vision–Language–Action Model framework • Research on multimodal learning and decision-making in contact-rich environments • Technical support for AI model adaptation and evaluation
Expected R&R of Korean SME Partner	• Provision of robotic platforms or application contexts for collaborative validation • Integration and testing of the proposed AI technology in practical environments • Joint exploration of application opportunities and commercialization pathways

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project is expected to contribute to the advancement of Physical AI technologies applicable to contact-rich robotic environments. Anticipated outcomes include improved adaptability of robotic systems, enhanced robustness in physical interaction tasks, and reduced reliance on rigid control strategies. From an industrial perspective, the technology offers potential value in manufacturing, assembly, and other domains where physical interaction plays a central role. The project also strengthens international collaboration between Korea and Canada in AI and robotics, fostering innovation and long-term industrial competitiveness.

6. References

(1) A. Brohan, N. Brown, J. Carbajal, *et al.*, *RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control*, *Proc. Conf. Robot Learn.* (CoRL), 2023.

(2) S. Stepputtis, J. Campbell, M. Phielipp, S. Lee, C. Baral, and H. Ben Amor, *Language-Conditioned Imitation Learning for Robot Manipulation Tasks*, *Adv. Neural Inf. Process. Syst.* **33**, 13104–13115 (2020).

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

(max 10 pages)